

PRACTICAL NO : 3

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Problem Statement: Circular Queue-- CO1, CO2

Implement Circular Queue using Array. Perform following operations on it. a. Insertion a fixed size of a queue.

Code :

```
#include <iostream>

using namespace std;

#define SIZE 5

class CircularQueue
{
    int queue[SIZE];
    int front, rear;

public:
    CircularQueue()
    {
        front = -1;
        rear = -1;
    }
    // Enqueue
    void enqueue(int value)
    {
        if ((rear + 1) % SIZE == front)
        {
            cout << "Queue is Full!" << endl;
```

```
    return;
}

if (front == -1)
    front = 0;

rear = (rear + 1) % SIZE;
queue[rear] = value;

cout << value << " inserted successfully." << endl;
}

// Dequeue
void dequeue()
{
    if (front == -1)
    {
        cout << "Queue is Empty!" << endl;
        return;
    }

    cout << queue[front] << " deleted Successfully." << endl;

    if (front == rear)
    {
        front = rear = -1;
    }

    else
    {
        front = (front + 1) % SIZE;
```

```

    }
}

// Display
void display()
{
    if (front == -1)
    {
        cout << "Queue is Empty!" << endl;
        return;
    }

    cout << "Circular Queue: ";

    int i = front;
    while (true)
    {
        cout << queue[i] << " ";

        if (i == rear)
            break;

        i = (i + 1) % SIZE;
    }

    cout << endl;
}
};

```

```
int main()
{
    CircularQueue q;
    int choice, value;

    do
    {
        cout << "\n----- Circular Queue Menu-----" << endl;
        cout << "1. Enqueue" << endl;
        cout << "2. Dequeue" << endl;
        cout << "3. Display" << endl;
        cout << "4. Exit" << endl;

        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1:
                cout << "Enter value: ";
                cin >> value;
                q.enqueue(value);
                break;
            case 2:
                q.dequeue();
                break;
            case 3:
                q.display();
```

```
        break;
    case 4:
        cout << "Program Exited." << endl;
        break;
    default:
        cout << "Invalid Choice!" << endl;
    }
} while (choice != 4);
return 0;
}
```

Output:

----- Circular Queue Menu -----

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 2

Queue is Empty!

----- Circular Queue Menu -----

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter value: 20

20 inserted successfully.

----- Circular Queue Menu -----

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 3

Circular Queue: 20

----- Circular Queue Menu -----

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 10

Invalid Choice!

----- Circular Queue Menu -----

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 4

Program Exited.

=== Code Execution Successful ===